AP Computer Science A: Lesson Plan

Topic: Deterministic Tracing, 1D Array Iteration & Error Taxonomy | Target: 45-Minute Period

PART 1: LESSON PLAN (COLLEGE BOARD CED ALIGNED)

Course & Target Level:	AP Computer Science A (Grades 10–12 / University Prep)
CED Unit & Topic Alignment:	Unit 4: Iteration (4.2) | Unit 6: Arrays (6.2 Traversing Arrays, 6.3 Array Algorithms)
Instructional Duration:	45 Minutes (Single Standard Period)
Core Pedagogical Focus:	Deterministic Memory Tracing, Bug Classification Taxonomy, AP FRQ Scoring Alignment

1. Enduring Understandings & Learning Objectives (College Board CED)

- **CON-2:** Programmers incorporate iteration, selection, and data structures to extend the range of problems that can be solved computationally.
- **CON-2.I:** Represent collections of related data using one-dimensional (1D) array objects.
- **CON-2.J:** Traverse the elements of a 1D array using iteration statements.
- **CON-2.J.1 (EK):** Iteration statements can be used to access all elements in an array (0 to array.length - 1).
- **CON-2.J.2 (EK):** An ArrayIndexOutOfBoundsException occurs when attempting to access an index outside the valid bounds [0, array.length - 1].
- **CON-2.I.3 (EK):** Array elements are accessed using the index operator []. Using loop index counter i without dereferencing accesses loop control state rather than stored memory content.

2. Measurable Behavioral Objectives (SWBAT)

- **1. Trace & State-Track:** Construct a manual 4-column trace table with 100% accuracy tracking loop control variables (i), array dereferences (arr[i]), and accumulator state changes (sum).
- **2. Error Taxonomy:** Systematically classify array traversal bugs into formal categories: (a) Index vs. Value Dereference Logic Error, (b) Incomplete Boundary Off-by-One, and (c) Terminal Runtime ArrayIndexOutOfBoundsException.
- **3. AP Rubric Alignment:** Evaluate code snippets against authentic College Board AP Free-Response Question (FRQ) Scoring Guidelines, identifying point loss traps and authoring compliant traversal headers.

3. Instructional Sequence & Lesson Timeline (45 Minutes)

Time	Instructional Phase & Teacher Script Actions	Active Student Learning & Check for Understanding
0:00–0:05 (5 min)	**Do Now & The Cognitive Trap** Display buggy loop (sum += i; with {3, 7, 2, 9}). Deliver hook: 'Syntax comes with practice, but top AP scores come from deterministic mental modeling. What does this print?'	Formulate independent hypothesis on handout. Identify the intuitive distractor (21) vs. actual trace output (6).
0:05–0:15 (10 min)	**Explicit Modeling: The 3 Core Traversal Bugs**	Active Annotation: Follow teacher modeling on Section 2 trace tables.

	Walk through trace table architecture. Introduce the Apartment Analogy: i is the apartment number; numbers[i] is the resident inside. Walk through: (1) Index vs Value, (2) Start Bound at 1, (3) Boundary breach <= length.	Annotate RAM memory cells and identify the exact moment ArrayIndexOutOfBoundsException throws.
0:15–0:28 (13 min)	**Differentiated Tiered Debugging (Pairs)** Circulate room guiding tiered inquiry. Tier 1 receives visual memory scaffolds; Tier 2 traces algorithmic searches; Tier 3 evaluates reverse-copy memory exceptions.	Pair Programming / Peer Diagnostics: - Tier 1: Fill trace tables with visual cells. - Tier 2: Debug min-value search algorithm. - Tier 3: Solve reverse-traversal boundary traps.
0:28–0:38 (10 min)	**AP FRQ Rubric Simulation** Display College Board FRQ 4-point array rubric. Present student code sample with traversal errors. Walk through how AP Readers deduct points for <= length vs < length.	Peer AP Reader Activity: Evaluate sample responses against the rubric checklist. Identify exact points awarded/deducted and author canonical corrections.
0:38–0:45 (7 min)	**Synthesis & Formative Exit Ticket** Consolidate the 3 Golden Traversal Invariants: Start at 0, strictly less than length, dereference with [i]. Administer independent exit check.	Independent Assessment: Complete 3-minute Exit Ticket tracing even-indexed accumulation without peer or reference support.

4. Pedagogical Scaffolding & Differentiation Architecture

- **Language & Conceptual Access:** Tier 1 (Scaffolded Support): Embedded sentence frames for formal technical justification; pre-drawn physical RAM memory diagrams; partially filled iteration trace state rows to minimize cognitive load on syntax while reinforcing indexing concepts.
- **Targeted Proficiency:** Tier 2 (Core Application): Standard algorithmic tracing without visual aids; independent identification of algorithmic logic bugs (e.g., storing the index rather than the value in a minimum-value search).
- **Advanced Synthesis:** Tier 3 (Mastery & Extension): Reverse-array traversal, complex nested dereferences, and mathematical index-mapping formulas (target[length - 1 - i]) mirroring high-rigor AP FRQ Part B prompts.